Mathematics from Zero for Computer Science Students

Numbers, Variables, Equations, Functions, Graphs, and Logic

Lecture notes for absolute beginners

A slow, practical introduction for students who need to rebuild the mathematical basics before calculus, algebra, geometry, programming, data analysis, or physics.

Generated lecture notes

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1 How to Use These Notes

These notes are written for students who do not yet feel safe with basic mathematics. The goal is not to impress anyone with formal definitions. The goal is to build working understanding.

The central path is:

$$\text{number} \longrightarrow \text{expression} \longrightarrow \text{equation} \longrightarrow \text{function} \longrightarrow \text{graph} \longrightarrow \text{algorithm}.$$

This path is especially natural for computer science students. In programming, you constantly use numbers, variables, expressions, conditions, functions, loops, and logical statements. These are mathematical ideas, even when they appear in code.

Key idea

A mathematical formula is often a very small program written in a compact notation.

For example,

$$f(x) = 2x + 3$$

means: take input x, multiply it by 2, add 3, and return the result.

In Python this is:

```
def f(x):
    return 2*x + 3
```

These notes repeat basic points many times on purpose. A student who is unsure about the meaning of $f(3)$, or about the difference between x and $3x$, cannot safely learn calculus, matrices, physics, machine learning, or numerical methods.

What this course tries to fix

Many mathematical failures are not caused by intelligence. They are caused by missing small steps. Typical missing steps are:

- not knowing the order of operations,
- not understanding that a variable can represent an unknown number or an input,
- confusing an expression with an equation,
- not understanding what $f(3)$ means,
- not knowing how to read a point on a graph,
- not knowing what an inequality describes,
- not seeing the connection between formulas and code.

This document rebuilds those steps slowly.

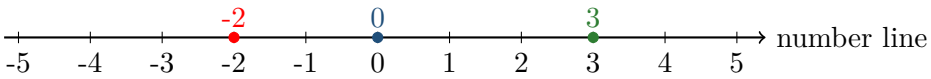
2 Numbers and Operations

2.1 Kinds of numbers

The most basic objects in mathematics are numbers. Different kinds of numbers are useful in different situations.

Set	Examples	Meaning
$\mathbb{N}$	$0, 1, 2, 3, \ldots$	natural numbers, used for counting
$\mathbb{Z}$	$\ldots, -2, -1, 0, 1, 2, \ldots$	integers
$\mathbb{Q}$	$\frac{1}{2}, -\frac{7}{3}, 0.25$	rational numbers, fractions of integers
$\mathbb{R}$	$\sqrt{2}, \pi, -3.7$	real numbers, points on the number line

For most of this course, we work with real numbers $\mathbb{R}$. This means we imagine a continuous number line.



2.2 Order of operations

Mathematical expressions must be read in a fixed order. The usual order is:

1. parentheses,
2. powers and roots,
3. multiplication and division,
4. addition and subtraction.

Example: order of operations

Calculate

$$2 + 3 \cdot 4.$$

Multiplication comes before addition, so

$$2 + 3 \cdot 4 = 2 + 12 = 14.$$

But

$$(2 + 3) \cdot 4 = 5 \cdot 4 = 20.$$

The parentheses changed the meaning.

Common mistake

Do not calculate from left to right blindly. The expression

$$2 + 3 \cdot 4$$

is not equal to $5 \cdot 4$. Multiplication is done first.

2.3 Fractions

A fraction is division written in another way:

$$\frac{a}{b} = a \div b, \qquad b \neq 0.$$

The denominator cannot be zero because division by zero has no meaning in ordinary arithmetic.

Example: adding fractions

Calculate

$$\frac{1}{2} + \frac{1}{3}.$$

We need a common denominator:

$$\frac{1}{2} = \frac{3}{6}, \qquad \frac{1}{3} = \frac{2}{6}.$$

Therefore

$$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}.$$

Common mistake

A very common wrong step is

$$\frac{1}{2} + \frac{1}{3} = \frac{1+1}{2+3} = \frac{2}{5}.$$

This is false. We do not add denominators like that.

2.4 Powers and roots

A power means repeated multiplication:

$$2^3 = 2 \cdot 2 \cdot 2 = 8.$$

A square root reverses squaring:

$$\sqrt{9} = 3, \qquad \text{because } 3^2 = 9.$$

Some useful rules are:

$$a^m a^n = a^{m+n}, \qquad \frac{a^m}{a^n} = a^{m-n}, \qquad (a^m)^n = a^{mn}.$$

Example: powers

Simplify

$$2^3 \cdot 2^4.$$

Since the base is the same, add exponents:

$$2^3 \cdot 2^4 = 2^{3+4} = 2^7 = 128.$$

2.5 Percentages

A percentage is a fraction with denominator 100:

$$15\% = \frac{15}{100} = 0.15.$$

If a price is 200 and increases by 15%, the increase is

$$0.15 \cdot 200 = 30.$$

The new price is

$$200 + 30 = 230.$$

Programming connection

Percentages often appear in data analysis, business calculations, error rates, accuracy, and memory usage.

```
price = 200
new_price = price * 1.15
```

Multiplying by 1.15 means adding 15%.

Practice

Calculate:

1. $3 + 4 \cdot 5$
2. $(3 + 4) \cdot 5$
3. $\frac{1}{4} + \frac{1}{2}$
4. 5^2
5. 20% of 80

3 Variables, Expressions, and Substitution

3.1 What is a variable?

A variable is a symbol that represents a value. The value may be unknown, changeable, or treated as input.

For example, in

$$x + 2,$$

the letter x represents a number. Until we know x, the expression $x + 2$ is not a single number.

Key idea

A variable is not decoration. It is a placeholder for a value.

In programming, variables are familiar:

```
x = 10
result = x + 2
```

Here x stores the value 10, so the result is 12.

3.2 Expression versus equation

An expression is something that can be evaluated if we know the values of the variables. Examples:

$$3x + 5, \qquad x^2 - 4, \qquad \frac{x+1}{2}.$$

An equation says that two expressions are equal. Examples:

$$3x + 5 = 20, \qquad x^2 - 4 = 0.$$

Object	Example	Meaning
Expression	$3x + 5$	can produce a value
Equation	$3x + 5 = 20$	asks when two sides are equal

3.3 Substitution

Substitution means replacing a variable by a number.

Example: substitution

Let

$$A(x) = 3x + 5.$$

If $x = 10$, then

$$A(10) = 3 \cdot 10 + 5 = 30 + 5 = 35.$$

Example: substitution with a negative number

Let

$$B(x) = x^2 + 2x.$$

Calculate $B(-3)$:

$$B(-3) = (-3)^2 + 2(-3) = 9 - 6 = 3.$$

Common mistake

When substituting a negative number, use parentheses. For example,

$$(-3)^2 = 9.$$

But

$$-3^2 = -(3^2) = -9.$$

These are different expressions.

3.4 Parameters

Sometimes a formula contains more than one symbol. For example,

$$f(x) = ax + b.$$

Here x is usually the input, while a and b are parameters. Parameters control the behavior of the function.

In code:

```
def f(x, a, b):
    return a*x + b
```

If $a = 2$ and $b = 3$, then

$$f(x) = 2x + 3.$$

Practice

Let

$$f(x) = 2x + 7, \qquad g(x) = x^2 - 1.$$

Calculate:

1. $f(0)$
2. $f(5)$
3. $g(3)$
4. $g(-2)$

4 Equations as Questions

4.1 What is an equation asking?

An equation is a question. For example,

$$2x + 3 = 11$$

means:

> For what value of x does the expression $2x + 3$ become equal to 11?

The answer is $x = 4$, because

$$2 \cdot 4 + 3 = 8 + 3 = 11.$$

Key idea

Solving an equation means finding an input that produces a required output.

4.2 Balance principle

An equation is like a balance. If two sides are equal, then doing the same valid operation to both sides keeps them equal.

From

$$2x + 3 = 11$$

subtract 3 from both sides:

$$2x = 8.$$

Divide both sides by 2:

$$x = 4.$$

Example: solving a linear equation

Solve

$$5x - 4 = 21.$$

Add 4 to both sides:

$$5x = 25.$$

Divide by 5:

$$x = 5.$$

Check:

$$5 \cdot 5 - 4 = 25 - 4 = 21.$$

4.3 Equations in programming terms

Suppose we have a function:

```
def f(x):
    return 2*x + 3
```

The equation

$$2x + 3 = 11$$

asks:

> Which input x makes `f(x)` return 11?

This is like debugging or searching for an input that gives a desired output.

4.4 Quadratic equations: the first idea

A quadratic equation contains x^2. Example:

$$x^2 = 9.$$

This equation has two solutions:

$$x = 3 \quad \text{or} \quad x = -3,$$

because

$$3^2 = 9, \qquad (-3)^2 = 9.$$

Common mistake

Do not forget the negative solution when solving $x^2 = a$. If $a > 0$, then usually

$$x = \sqrt{a} \quad \text{or} \quad x = -\sqrt{a}.$$

Example: simple quadratic equation

Solve

$$x^2 - 4 = 0.$$

Move -4 to the other side:

$$x^2 = 4.$$

Therefore

$$x = 2 \quad \text{or} \quad x = -2.$$

Practice

Solve and check:

1. $x + 5 = 12$
2. $3x = 18$
3. $4x - 1 = 15$
4. $x^2 = 25$
5. $x^2 - 9 = 0$

5 Inequalities and Conditions

5.1 Meaning of inequalities

An inequality compares values:

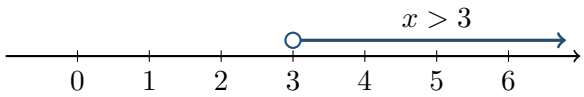
$$x > 3, \qquad x < 10, \qquad x \geq 0, \qquad x \leq 100.$$

These statements describe sets of numbers.

For example,

$$x > 3$$

means all numbers greater than 3: 4, 5, 3.1, 100, and infinitely many more.



The empty circle at 3 means that 3 is not included.

5.2 Closed and open intervals

Intervals are compact ways to describe ranges.

Notation	Meaning	Includes endpoints?
$[0, 1]$	$0 \leq x \leq 1$	both endpoints
$(0, 1)$	$0 < x < 1$	neither endpoint
$[0, 1)$	$0 \leq x < 1$	left endpoint only
$(0, 1]$	$0 < x \leq 1$	right endpoint only

5.3 Conditions in code

Inequalities are the mathematics behind `if` statements.

```
if x > 10:
    print("large")
else:
    print("small")
```

The condition $x > 10$ is either true or false.

A double inequality such as

$$0 \leq x \leq 100$$

means that x must be at least 0 and at most 100.

In Python:

```
if 0 <= x <= 100:
    print("valid percentage")
```

5.4 Solving simple inequalities

Solving inequalities is similar to solving equations, but with one crucial rule: multiplying or dividing by a negative number reverses the inequality sign.

Example: linear inequality

Solve

$$2x + 3 < 11.$$

Subtract 3:

$$2x < 8.$$

Divide by 2:

$$x < 4.$$

Example: negative coefficient

Solve

$$-2x < 8.$$

Divide by -2. The sign changes:

$$x > -4.$$

Common mistake

When dividing an inequality by a negative number, reverse the sign:

$$-2x < 8 \quad \Rightarrow \quad x > -4.$$

Not $x < -4$.

6 Coordinates and the Cartesian Plane

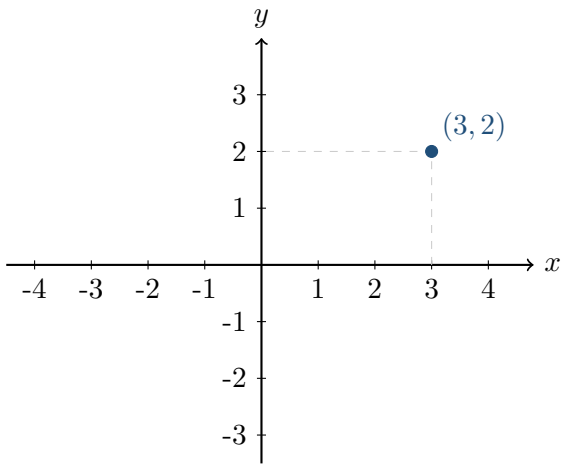
6.1 From a number line to a plane

A number line uses one coordinate. A plane uses two coordinates.

A point in the plane is written as

$$(x, y).$$

The first coordinate tells us horizontal position. The second coordinate tells us vertical position.



Key idea

The point $(3, 2)$ means: move 3 units right and 2 units up.

6.2 Reading points

The point $(-2, 4)$ means:

- $x = -2$: move left by 2,
- $y = 4$: move up by 4.

The point $(5, -1)$ means:

- $x = 5$: move right by 5,
- $y = -1$: move down by 1.

6.3 Distance on a line

The distance between two numbers a and b on the number line is

$$|a - b|.$$

For example, the distance between -2 and 5 is

$$|5 - (-2)| = |7| = 7.$$

6.4 Distance in the plane

For two points

$$A = (x_1, y_1), \qquad B = (x_2, y_2),$$

the distance is

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

This is the Pythagorean theorem.

Example: distance between points

Find the distance between

$$A = (1, 2), \qquad B = (4, 6).$$

The horizontal change is

$$4 - 1 = 3.$$

The vertical change is

$$6 - 2 = 4.$$

Therefore

$$d(A, B) = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

Programming connection

Coordinates are everywhere in computer science: screen pixels, graphics, games, robotics, plots, maps, arrays, image processing, and geometry algorithms.

Practice

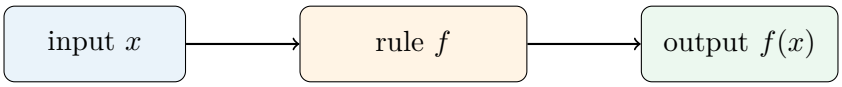
For the points $A = (2, 1)$, $B = (6, 4)$, $C = (-1, 3)$:

1. Draw them on a coordinate plane.
2. Find the distance between A and B.
3. Which point has negative x-coordinate?

7 Functions as Input-Output Machines

7.1 The basic idea

A function takes an input and produces an output according to a rule.



For example:

$$f(x) = 2x + 3.$$

If the input is $x = 5$, then

$$f(5) = 2 \cdot 5 + 3 = 13.$$

7.2 Function notation

The notation

$$f(5)$$

does not mean $f \cdot 5$. It means: apply the function f to the input 5.

Common mistake

Do not read $f(5)$ as multiplication. In function notation, parentheses mean input.

7.3 Functions and code

The mathematical function

$$f(x) = 2x + 3$$

corresponds to the Python function:

```
def f(x):
    return 2*x + 3
```

The expression $f(5)$ corresponds to:

```
f(5)
```

which returns 13.

7.4 Tables of values

A table helps us understand a function by testing inputs.

For

$$f(x) = x^2,$$

we get:

x	-2	-1	0	1	2
$f(x) = x^2$	4	1	0	1	4

7.5 Domain and output values

The domain is the set of allowed inputs. For example,

$$f(x) = \frac{1}{x}$$

cannot use $x = 0$, because division by zero is not allowed. Therefore the domain is all real numbers except 0:

$$x \in \mathbb{R}, \quad x \neq 0.$$

Example: allowed input

For

$$g(x) = \sqrt{x},$$

ordinary real-number mathematics requires

$$x \geq 0.$$

So $g(9) = 3$, but $g(-1)$ is not a real number.

Practice

Let

$$f(x) = 3x - 2, \qquad h(x) = \frac{10}{x}.$$

Calculate:

1. $f(0)$
2. $f(4)$
3. $h(2)$
4. $h(5)$
5. Why is $h(0)$ not allowed?

8 Graphs: Pictures of Functions

8.1 What a graph means

A graph of a function $y = f(x)$ is a set of points

$$(x, f(x)).$$

This means: for each allowed input x, calculate the output $f(x)$, then draw the point $(x, f(x))$.

Key idea

If the point $(3, 7)$ is on the graph of $y = f(x)$, then $f(3) = 7$.

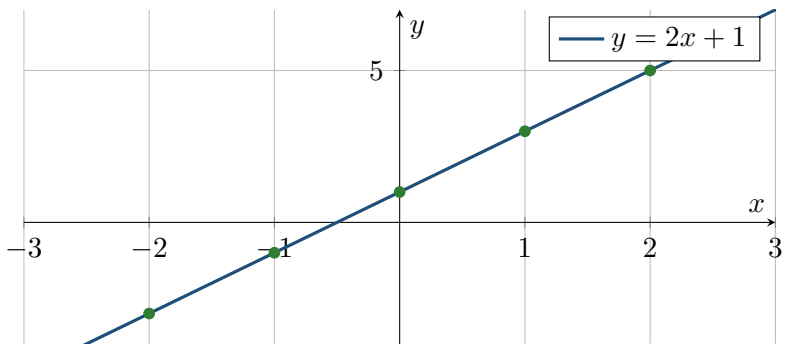
8.2 Graph of a linear function

Consider

$$f(x) = 2x + 1.$$

Some values are:

x	-2	-1	0	1	2
$f(x)$	-3	-1	1	3	5

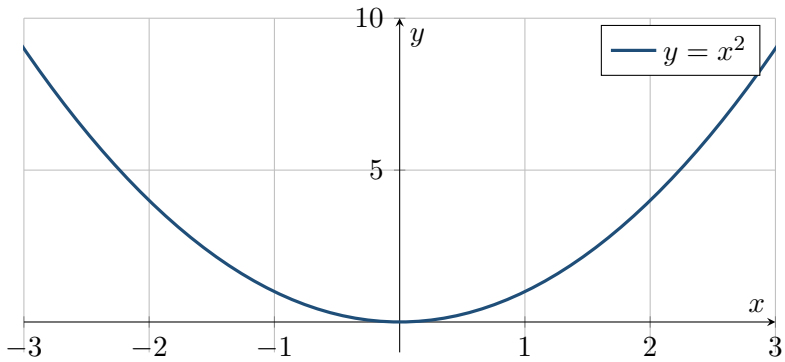


8.3 Graph of a quadratic function

The function

$$f(x) = x^2$$

has a U-shaped graph called a parabola.



Negative and positive inputs can give the same output:

$$(-2)^2 = 4, \qquad 2^2 = 4.$$

8.4 Intersections with axes

An intersection with the y-axis happens when $x = 0$. For

$$f(x) = 2x + 1,$$

we get

$$f(0) = 1,$$

so the graph crosses the y-axis at $(0, 1)$.

An intersection with the x-axis happens when $y = 0$. To find it, solve

$$f(x) = 0.$$

For $f(x) = 2x + 1$:

$$2x + 1 = 0, \quad 2x = -1, \quad x = -\frac{1}{2}.$$

So the graph crosses the x-axis at $(-\frac{1}{2}, 0)$.

Common mistake

The x-axis crossing is found by setting $y = 0$, not by setting $x = 0$. The y-axis crossing is found by setting $x = 0$.

9 Linear Functions

9.1 The form $f(x) = ax + b$

A linear function has the form

$$f(x) = ax + b.$$

The number a controls how fast the function changes. The number b is the output when $x = 0$.

Parameter	Meaning	Graph effect
a	slope / rate of change	makes the line steeper or flatter
b	intercept / starting value	moves the line up or down

9.2 Slope as change in output per input

For a line, the slope is

$$a = \frac{\Delta y}{\Delta x}.$$

This means:

$$\text{slope} = \frac{\text{change in output}}{\text{change in input}}.$$

Example: cloud cost model

Suppose a cloud service costs 10 units fixed plus 0.05 units per request. If n is the number of requests, then

$$C(n) = 10 + 0.05n.$$

Here:

- 10 is the fixed cost,
- 0.05 is the cost per request,
- $C(n)$ is the total cost.

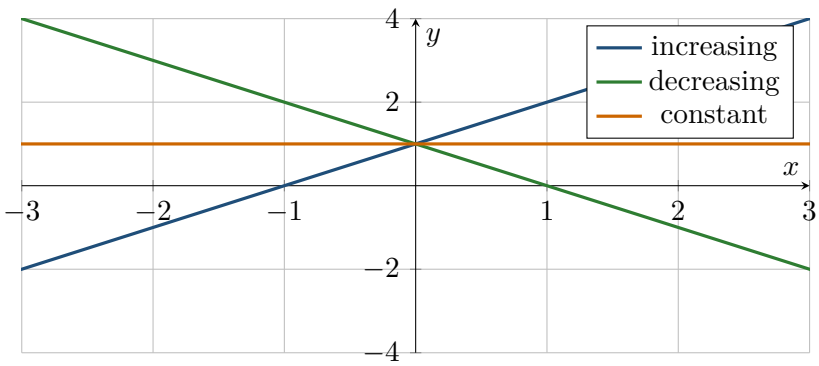
If $n = 1000$, then

$$C(1000) = 10 + 0.05 \cdot 1000 = 10 + 50 = 60.$$

9.3 Increasing, decreasing, constant

For $f(x) = ax + b$:

- if $a > 0$, the function increases,
- if $a < 0$, the function decreases,
- if $a = 0$, the function is constant.



Practice

For each function, identify a, b, and say whether the function is increasing, decreasing, or constant:

1. $f(x) = 3x + 1$
2. $g(x) = -2x + 5$
3. $h(x) = 7$

10 Quadratic Functions

10.1 The basic quadratic function

The simplest quadratic function is

$$f(x) = x^2.$$

Its graph is a parabola. The output is never negative, because a square is nonnegative:

$$x^2 \geq 0.$$

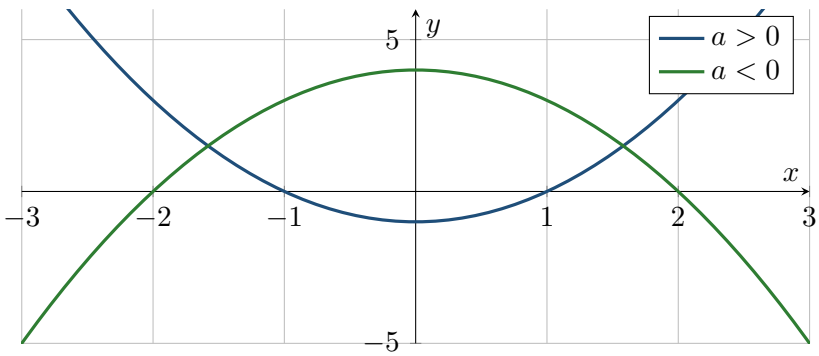
10.2 The general form

A quadratic function often appears as

$$f(x) = ax^2 + bx + c, \qquad a \neq 0.$$

The coefficient a controls whether the parabola opens upward or downward:

- if $a > 0$, it opens upward,
- if $a < 0$, it opens downward.



10.3 Zeros of a quadratic function

Zeros are inputs for which the output is zero. They solve

$$f(x) = 0.$$

For example,

$$f(x) = x^2 - 4.$$

Zeros solve

$$x^2 - 4 = 0.$$

Therefore

$$x^2 = 4,$$

so

$$x = -2 \quad \text{or} \quad x = 2.$$

The graph crosses the x-axis at $(-2, 0)$ and $(2, 0)$.

Example: quadratic model

Suppose a simple performance model is

$$T(n) = n^2 + 3n + 2.$$

This is not a precise runtime model for every program, but it shows the idea that quadratic growth becomes much larger than linear growth for large n.
For $n = 10$:

$$T(10) = 100 + 30 + 2 = 132.$$

For $n = 100$:

$$T(100) = 10000 + 300 + 2 = 10302.$$

The quadratic term dominates.

11 Powers, Roots, Exponentials, and Logarithms

11.1 Power functions

Functions such as

$$f(x) = x, \qquad f(x) = x^2, \qquad f(x) = x^3$$

are power functions.

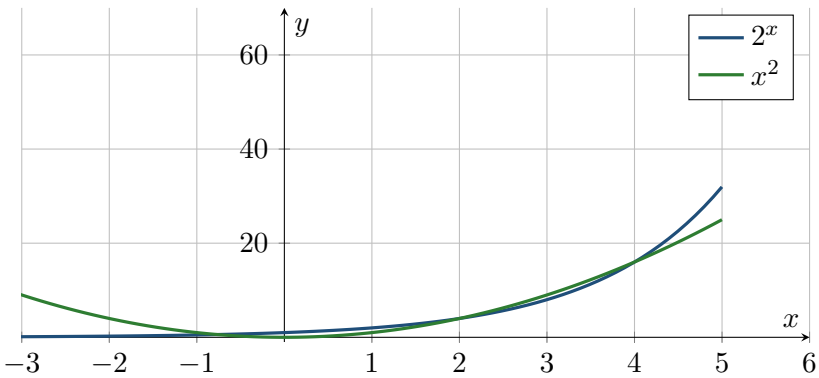
They are important in algorithm analysis because we compare growth rates.

11.2 Exponential functions

An exponential function has the variable in the exponent:

$$f(x) = 2^x.$$

It grows very fast.



For small x, exponential growth may not look terrifying. For larger x, it becomes enormous.

11.3 Logarithms

A logarithm answers the question: what exponent is needed?

$$\log_2(8) = 3$$

because

$$2^3 = 8.$$

So logarithms reverse exponentials.

Statement	Equivalent statement	Meaning
$\log_2(8) = 3$	$2^3 = 8$	exponent needed is 3
$\log_{10}(100) = 2$	$10^2 = 100$	exponent needed is 2
$\log_2(1) = 0$	$2^0 = 1$	exponent needed is 0

Programming connection

Logarithms appear in binary search, balanced trees, information theory, complexity analysis, compression, entropy, and machine learning loss functions.

11.4 Growth rates

For computer science, the rough order of growth matters:

$$\log n \quad < \quad n \quad < \quad n \log n \quad < \quad n^2 \quad < \quad 2^n.$$

This is not about exact values for tiny inputs. It is about what happens as n becomes large.

Example: binary search

If a sorted array has about $1,000,000$ elements, binary search needs about

$$\log_2(1,000,000) \approx 20$$

steps, not one million steps. That is why logarithmic algorithms are powerful.

12 Sequences as Generated Lists

12.1 What is a sequence?

A sequence is an ordered list of numbers:

$$a_1, a_2, a_3, \ldots$$

A formula may define the n-th element. For example,

$$a_n = 2n$$

gives

$$2, 4, 6, 8, 10, \ldots$$

Programming connection

A sequence is like a list generated by a rule.

```
def a(n):
    return 2*n

values = [a(n) for n in range(1, 6)]
# [2, 4, 6, 8, 10]
```

12.2 Arithmetic sequences

An arithmetic sequence adds the same number each time.

Example:

$$3, 7, 11, 15, 19, \ldots$$

Here the difference is 4.

A formula is

$$a_n = 3 + 4(n - 1).$$

12.3 Geometric sequences

A geometric sequence multiplies by the same number each time.

Example:

$$2, 6, 18, 54, \ldots$$

Here the ratio is 3.

A formula is

$$a_n = 2 \cdot 3^{n-1}.$$

12.4 Recursive definitions

A recursive definition gives a starting value and a rule for the next value.

Example:

$$a_1 = 1, \qquad a_{n+1} = a_n + 2.$$

This generates

$$1, 3, 5, 7, 9, \ldots$$

In Python:

```
a = 1
for n in range(5):
    print(a)
    a = a + 2
```

Practice

Find the first five terms:

1. $a_n = n^2$
2. $b_n = 3n + 1$

3. $c_1 = 2,\ c_{n+1} = 2c_n$

13 Basic Logic for Mathematics and Programming

13.1 Statements and truth values

A mathematical statement is a sentence that is true or false.

Examples:

- $2 + 2 = 4$ is true.
- $5 < 3$ is false.
- $x + 1$ is not a statement, because it is just an expression.

13.2 AND, OR, NOT

Logic uses operations on truth values.

Logic	Mathematics	Python
and	$P \wedge Q$	`P and Q`
or	$P \vee Q$	`P or Q`
not	$\neg P$	`not P`

Example: AND condition

The condition

$$0 \leq x \leq 100$$

means

$$x \geq 0 \quad \text{and} \quad x \leq 100.$$

Both parts must be true.

13.3 Implication

The statement

$$P \Rightarrow Q$$

means: if P is true, then Q is true.

Example:

> If a number is divisible by 4, then it is even.

This is true. But the reverse is false:

> If a number is even, then it is divisible by 4.

Counterexample: 6 is even, but 6 is not divisible by 4.

Key idea

A counterexample is one example that proves a universal statement false.

13.4 Necessary and sufficient conditions

A condition is sufficient if it guarantees the result.

Being divisible by 4 is sufficient for being even.

A condition is necessary if the result cannot happen without it.

Being even is necessary for being divisible by 4, because every number divisible by 4 is even.

Common mistake

Students often confuse a statement with its converse. The statement

$$P \Rightarrow Q$$

is not the same as

$$Q \Rightarrow P.$$

14 From Formula to Algorithm

14.1 A formula as a deterministic procedure

A formula gives instructions. For example,

$$f(x) = 3x^2 - 2x + 1$$

means:

1. take x,
2. square it,
3. multiply by 3,
4. subtract $2x$,
5. add 1.

In code:

```
def f(x):
    return 3*x**2 - 2*x + 1
```

14.2 Tables by loops

A table of values can be produced by a loop.

```
def f(x):
    return 2*x + 1

for x in [-2, -1, 0, 1, 2]:
    print(x, f(x))
```

This corresponds to the mathematical table:

x	-2	-1	0	1	2
$f(x)$	-3	-1	1	3	5

14.3 Searching for a solution

The equation

$$f(x) = 0$$

asks for an input where the output is zero. A computer can sometimes search for such an input.

For example:

```
def f(x):
    return x*x - 4

for x in range(-5, 6):
    if f(x) == 0:
        print("solution", x)
```

This finds $x = -2$ and $x = 2$.

14.4 Discrete versus continuous

A computer usually checks discrete values: $-5, -4, -3, \ldots, 5$. Mathematics often studies continuous values, such as all real numbers in an interval.

Type	Example	Interpretation
Discrete	$n \in \mathbb{N}$	separate values, like list indices
Continuous	$x \in \mathbb{R}$	every point on an interval

This difference is important in numerical methods, simulations, graphics, signals, and machine learning.

15 Mini-Projects for Computer Science Students

15.1 Project 1: Plot a function

Choose

$$f(x) = x^2 - 4.$$

Make a table for $x = -4, -3, \ldots, 4$, then draw the graph. Mark the zeros.

Expected zeros:

$$x = -2, \qquad x = 2.$$

15.2 Project 2: Cost function

A service has fixed cost 50 and cost 0.02 per request. Write a function for total cost:

$$C(n) = 50 + 0.02n.$$

Answer:

1. What is $C(1000)$?
2. How many requests give cost 150?

Solution for the second question:

$$50 + 0.02n = 150.$$

Subtract 50:

$$0.02n = 100.$$

Divide by 0.02:

$$n = 5000.$$

15.3 Project 3: Valid input range

A percentage value p must satisfy

$$0 \leq p \leq 100.$$

Write the equivalent Python condition.

Answer:

```
if 0 <= p <= 100:
    print("valid")
else:
    print("invalid")
```

15.4 Project 4: Sequence generator

The sequence

$$a_n = 3n + 2$$

models generated values. Find the first ten terms and write a loop that prints them.

```
for n in range(1, 11):
    print(3*n + 2)
```

16 Common Mistakes and How to Repair Them

16.1 Mistake: treating all parentheses as multiplication

$$f(3)$$

means function input, not multiplication.

But

$$3(4 + 2)$$

means multiplication:

$$3(4+2) = 18.$$

Context matters.

16.2 Mistake: forgetting parentheses around negative numbers

Correct:

$$(-2)^2 = 4.$$

Different expression:

$$-2^2 = -4.$$

16.3 Mistake: adding fractions incorrectly

Wrong:

$$\frac{1}{2} + \frac{1}{3} = \frac{2}{5}.$$

Correct:

$$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}.$$

16.4 Mistake: thinking $x^2 = 9$ has only one solution

Correct:

$$x^2 = 9 \quad \Rightarrow \quad x = 3 \text{ or } x = -3.$$

16.5 Mistake: confusing axis intersections

For $y = f(x)$:

- y-axis: set $x = 0$,
- x-axis: set $y = 0$, which means solve $f(x) = 0$.

16.6 Mistake: confusing implication with converse

True:

$$\text{divisible by } 4 \Rightarrow \text{even}.$$

False converse:

$$\text{even} \Rightarrow \text{divisible by } 4.$$

Counterexample: 6.

17 Exercises with Selected Solutions

Exercise set A: arithmetic

1. Calculate $6 + 2 \cdot 5$.
2. Calculate $(6 + 2) \cdot 5$.
3. Calculate $\frac{2}{3} + \frac{1}{6}$.

4. Calculate 3^4.
5. Find 12% of 250.

Solutions A

1. $6 + 2 \cdot 5 = 6 + 10 = 16$.
2. $(6 + 2) \cdot 5 = 8 \cdot 5 = 40$.
3. $\frac{2}{3} + \frac{1}{6} = \frac{4}{6} + \frac{1}{6} = \frac{5}{6}$.
4. $3^4 = 81$.
5. $0.12 \cdot 250 = 30$.

Exercise set B: substitution

Let

$$f(x) = x^2 + 3x - 1.$$

Calculate:

1. $f(0)$
2. $f(2)$
3. $f(-1)$

Solutions B

1. $f(0) = 0^2 + 3 \cdot 0 - 1 = -1$.
2. $f(2) = 2^2 + 3 \cdot 2 - 1 = 4 + 6 - 1 = 9$.
3. $f(-1) = (-1)^2 + 3(-1) - 1 = 1 - 3 - 1 = -3$.

Exercise set C: equations

Solve:

1. $x - 7 = 3$
2. $2x + 5 = 17$
3. $-3x = 12$
4. $x^2 = 16$
5. $x^2 - 1 = 0$

Solutions C

1. $x = 10$.
2. $2x = 12$, so $x = 6$.
3. $x = -4$.
4. $x = 4$ or $x = -4$.
5. $x^2 = 1$, so $x = 1$ or $x = -1$.

Exercise set D: functions and graphs

For

$$f(x) = -x + 4,$$

answer:

1. Is the function increasing or decreasing?
2. What is $f(0)$?
3. Where does it cross the x-axis?

Solutions D

1. It is decreasing because the slope is $-1 < 0$.
2. $f(0) = 4$.
3. Set $f(x) = 0$:

$$-x + 4 = 0 \Rightarrow x = 4.$$

   The crossing is $(4, 0)$.

Exercise set E: logic

Decide whether each statement is true or false:

1. If a number is divisible by 10, then it is divisible by 5.
2. If a number is divisible by 5, then it is divisible by 10.
3. If $x > 5$, then $x > 2$.
4. If $x > 2$, then $x > 5$.

Solutions E

1. True.
2. False. Counterexample: 15.
3. True.
4. False. Counterexample: 3.

18 Cheat Sheet

Summary

Order of operations:

$$\text{parentheses} \to \text{powers} \to \text{multiplication/division} \to \text{addition/subtraction}.$$

Fractions:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \qquad b, d \neq 0.$$

Linear function:

$$f(x) = ax + b.$$

a is slope. $b = f(0)$ is the y-intercept.

Function notation:

$$f(3) \text{ means input 3 into } f.$$

Graph of a function:

$$\text{points are } (x, f(x)).$$

If $(3, 7)$ is on the graph of f, then $f(3) = 7$.
Solving an equation: do the same valid operation to both sides.
Solving $x^2 = a$, $a > 0$:

$$x = \sqrt{a} \quad \text{or} \quad x = -\sqrt{a}.$$

Inequality rule: multiplying or dividing by a negative number reverses the sign.
Coordinates:

$$(x, y) = \text{horizontal coordinate, vertical coordinate.}$$

Distance between points:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Logarithm:

$$\log_b(a) = c \quad \Longleftrightarrow \quad b^c = a.$$

Logic:

$$P \Rightarrow Q \text{ is not the same as } Q \Rightarrow P.$$

19 Suggested Next Step

After students understand this material, the next lecture notes can safely introduce:

1. algebra from zero,
2. analytic geometry from zero,
3. calculus from zero,
4. matrices from zero,
5. physics from zero,
6. numerical methods and simulations.

The most important prerequisite is not memorizing formulas. It is understanding what a variable is, what a function does, and how a graph represents input-output behavior.